When AI Can't Compute: The Unbridgeable Gap in AI Solvability

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***Abstract*—Artificial Intelligence (AI) works well for many tasks like recognizing images, understanding language, and making decisions. However, AI can't solve every problem due to computing limitations and complex situations. This paper tackles an important question: How can we tell if AI can actually solve a specific problem before we try? We created SOLVE-X, a new method that helps determine whether a problem can be solved by AI or not. Our approach combines practical measures (like how difficult a problem is to compute and how much human context matters) with mathematical principles that show some problems are simply impossible to solve completely. SOLVE-X works like a systematic checklist that gives a clear yes/no answer about whether AI can handle a particular problem. We tested it on real-world examples from different fields. Unlike other methods that just guess based on past experience, SOLVE-X can actually prove when a problem cannot be solved by AI, giving researchers confidence in their conclusions. Our tests show that SOLVE-X is particularly good at avoiding false hopes (saying AI can solve something when it can't) and handling problems involving ethics and complex human situations. This tool can help AI researchers and practitioners make smarter decisions about when and where to use AI, leading to more thoughtful and responsible AI applications.**

Keywords*—*Artificial Intelligence (AI), algorithm, P and NP Problems, Explainable AI, Turing machine, AI-Completeness, radar plots.

I. Introduction

Artificial Intelligence (AI) has come to be a central element of modern technological systems, widely utilized across a range of areas to solve advanced computational and decision-making problems. From optimization methods that calculate cost-effective routing solutions to foreseen models that estimate educational performance results, AI systems are increasingly used to solve problems that have hitherto been regarded as outside computational capabilities. For instance, AI has been successfully deployed in precision agriculture, where deep learning models control irrigation valves based on soil moisture and temperature patterns, improving sustainability and resource efficiency [1]. Recent breakthroughs have even shown AI's ability to learn machine learning models for forecasting project duration using categorical features and project details [2]. Such increased deployment of AI is a wider paradigmatic realignment toward computational methods for resolving not just technical optimization issues but also sophisticated challenges with human cognition and contextual decision-making mechanisms.

Yet this extensive dependence on AI systems calls for careful scrutiny of their intrinsic capabilities and limits. The core research question arises: Are there well-defined limits to AI's problem-solving power, and are these limits formally definable? Although AI has shown impressive progress in pattern recognition, strategic game-playing, and natural language processing, it is still bounded by inherent restrictions—computational and theoretical—separating AI-solvable problems from those outside its ability. This problem is encapsulated by the notion of AI-Completeness, which formalizes that there are problems solvable only by human-level intelligence [3]. This is analogous to the measurement complexity in the Turing Test, when AI systems can output approximate answers without theoretical assurances of correctness.

The computational complexity theory has classically divided problems into Polynomial Time (P) and Non-Polynomial Time (NP) classes [4]. Classical forecasting tasks like time-series prediction have long used ARIMA and other statistical models where solvability is well-defined and predictable [5]. The shift towards AI for similar problems requires an updated evaluation lens. Although this taxonomy presents a formal basis for algorithmic analysis, the question is whether similar formal criteria exist for measuring AI solvability. In particular, can a scientific methodology be established that yields deterministic classification of whether a problem at hand is within AI's computational capabilities or beyond its theoretical limitations?

Meta's AI content moderation tools exemplify the challenges of applying AI to contextually rich real-world issues [6]. While effective at scale, these systems struggle with content requiring nuanced understanding, cultural awareness, and low-resource language analysis. Leung (2022) highlights systemic "shortcuts and shortfalls," such as over-enforcing harmless content while under-enforcing harmful material [6]. The Rohingya crisis further reveals limitations in addressing hate speech, underscoring AI's difficulties with contextual depth, ethical judgment, and cultural sensitivity [7]. Similarly, AI-driven forest fire detection systems grapple with balancing false alarms and timely alerts, reflecting broader contextual and ethical challenges [8].

The paper tackles the question of how to assess whether AI can be used to address particular problems through presenting SOLVE-X. While empirical approaches rely on practical examples, SOLVE-X employs theoretical criteria—such as computational complexity and formal constraints—to categorize problems deterministically.

Through examples and comparisons, SOLVE-X illustrates how it bridges the gap of prevailing frameworks and allows for more informed, ethical, and pragmatic decisions regarding deploying AI. This fills the gap between computational theory and actual AI evaluation, enabling more trustworthy and contextual solvability analysis.

II. Related Work

A number of newer methodologies have been recently developed to solve the problem of assessing AI problem solvability, each with differing solutions to this basic question. Although these do not really solve our problem, they work on a similar principle and help to tackle it to a great extent.

A. ADeLe Framework (Microsoft Research)

The ADeLe Framework addresses the AI solvability issue by a shift away from the conventional task-level evaluation and towards a holistic, ability-driven evaluation framework [9]. Rather than offering simplistic binary solvable/unsolvable labels, ADeLe decomposes AI competencies into 18 distinct rubrics, thereby creating an expansive evaluation space. ADeLe addresses the solvability challenge by understanding that various problems call for various combinations of abilities. For example, a math problem might require logical reasoning and numerical computation, whereas a creative writing prompt needs linguistic skills and creativity. By mapping tasks to the needed ability profile and evaluating them against an AI system's established skills across these 18 rubrics, ADeLe allows for more precise predictions regarding solvability, revolutionizing AI evaluation with 88% prediction accuracy. This allows researchers to pinpoint the very reasons for the long-standing hardness of certain problems and to identify particular capabilities that require enhancement.

B. MCP-Radar: Tool-Augmented Evaluation

MCP-Radar tackles the problem of solvability by accepting that modern AI systems more and more act as coordinators of external tools and less and less as standalone problem solvers [10]. The method systematically breaks down problem-solving processes into two independent aspects: the capabilities inherent in the large language model (LLM) and the capabilities outsourced to special-purpose tools. For example, in evaluating coding problems, MCP-Radar separates the reasoning and planning capabilities provided by the LLM (e.g., requirement understanding and algorithm planning) from computational execution done by code interpreters or debuggers. In domains such as service discovery and clustering, contextual language models like SBERT have shown that effective task solvability depends on representational richness and semantic granularity [11]. This separation enables more knowledgeable solvability decisions by enabling the independent examination of the sufficiency of available tools and the LLM's ability to effectively coordinate their use. The method accepts that solvability in modern AI systems is not only dependent on the model's inherent capabilities but also on the overall tooling environment and the model's ability to select and coordinate tools.

C. WildBench: Real-World Validation

WildBench measures solvability by grounding judgments in real user interactions rather than in simulated benchmarks [12]. WildBench is composed of 1,024 carefully selected tasks based on more than one million real human-chatbot conversation logs. These provide unique insight into users' challenges in real-world contexts. WildBench eliminates a major deficiency of conventional solvability testing: the gap between performance on controlled benchmarks and performance on actual problem-solving contexts. WildBench tasks often include incomplete specifications, implicit requirements, and contextual nuances that strongly affect solvability. By focusing its attention on challenges that intrinsically result from user interactions across a range of domains, including writing, analysis, reasoning, and complex problem-solving, WildBench provides a more ecologically valid measure of AI solvability. The system's focus on real-world complexity allows the detection of instances where the AI system will fail on tasks that appear straightforward in controlled environments but become complex due to vagueness and context-sensitivity of requirements.

D. Stanford HELM: Full Benchmarking

The Holistic Evaluation of Language Models (HELM) project addresses solvability through systematic evaluation across more than 30 diverse scenarios across question answering, safety, logical reasoning, and numerous other cognitive tasks [13]. HELM's solvability evaluation method is characterized by its scope and standardized process, enabling full performance data across problem domains. The model allows for domain-specific pattern detection in strengths and weaknesses of AI via extensive empirical evaluation. While HELM does not predict solvability for an individual problem, its full coverage has the empirical foundation for AI problem-solving bounds understanding. The systematic nature of the model allows researchers to identify which types of problems reliably challenge AI systems and which domains demonstrate reliable success, resulting in wider understanding of patterns of solvability across problem classes.

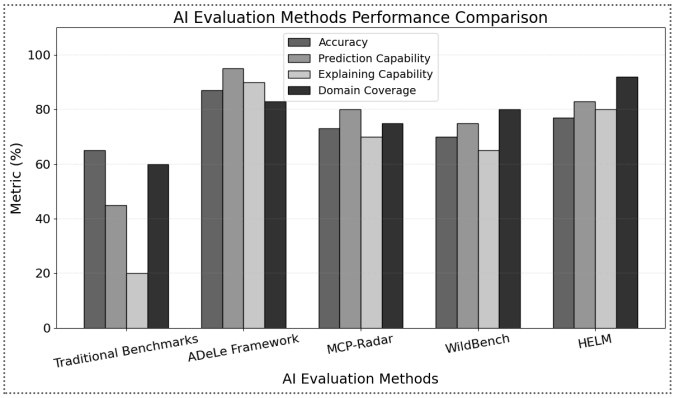


Fig. 1. Comparative performance of AI evaluation frameworks across key metrics (Accuracy, Prediction Capability, Explanation Capability, and Domain Coverage). Traditional benchmarks are contrasted with modern approaches (ADeLe, MCP-Radar, WildBench, HELM), with normalized scores (%) illustrating relative strengths.

Fig. 1 shows a general comparison among the existing radar plot models for assessing AI solvability. The methods outlined together capture various philosophical perspectives of measuring AI solvability: ability-based decomposition (ADeLe), tool-integration analysis (MCP-Radar), real-world verification (WildBench), and end-to-end empirical benchmarking (HELM). Beyond general-purpose AI benchmarking, domain-specific applications of AI have yielded optimization frameworks in networking, such as service function chaining using NFV, which demonstrates the utility of intelligent models in highly constrained and dynamic environments [14]. Each model captures distinct insights to advance the understanding of the conditions under which AI systems can effectively address particular problems, thus advancing the discipline towards higher-level and more precise solvability predictions.

III. Literature Review: Challenges faced

The assessment of AI problem solvability faces two fundamental challenges: the inherent opacity of modern AI systems, where decision-making processes function as unexplainable "black boxes" [15], and the existence of mathematically proven limitations that establish entire classes of problems as algorithmically unsolvable [16]. Together, these issues—one stemming from empirical system design and the other from deep theoretical constraints—create persistent obstacles in defining the boundaries of AI capabilities. Explainable AI (XAI) tools like LIME have also been used in medical diagnostics to provide interpretability in predictions, which may help estimate solvability by clarifying feature contributions to outcomes [17]. This section systematically examines four such barriers through formal computational frameworks (Table I), analyzing their consequences for solvability assessment.

A. Turing's Halting Problem: The Undecidability Barrier

Formalization: Let HP*(M,w)* represent whether Turing machine *M* halts on input *w*. Turing proved that HP is undecidable: no algorithm exists that outputs, HP(*M*,*w*) $\forall$(*M*,*w*) [18]. This implies:

$$\nexists\, A_{AI} : A_{AI}(P)\ solves\ P\ if\ P \equiv HP$$

AI-Constrained Domains: Self-referential systems, logical consistency verification, and solvability meta-problems. Content moderation of culturally nuanced hate speech exemplifies such undecidable contexts.

B. Gödel's Incompleteness Theorems: Provability Limits

Formalization: For any consistent formal system *F* capable of basic arithmetic:

1. First Theorem:

$$\exists\ statement\ G:\ F \nvdash G \wedge F \nvdash \neg G$$

2. Second Theorem:

$$F \nvdash Consis(F)$$

This imposes:

$$AT(P) = 0\ if\ P \equiv Provability(G)$$

AI Impact: AI systems manipulating formal proofs encounter true-but-unprovable statements, limiting automated theorem proving in mathematical logic [19].

C. Church-Turing Thesis: Computability Boundaries

Formalization: All effectively computable functions f satisfy:

$$f: N^k \to N \Leftrightarrow \exists\, TM\ M : M\ computes\ f$$

Corollary: Problems requiring non-algorithmic human cognition (e.g., ethical dilemmas or creative abstraction) violate this equivalence [20]:

$$P_{creative} \notin TM - Computable \Rightarrow AT(P_{creative}) = 0$$

Example: Generating truly novel mathematical concepts (e.g., AlphaTensor's matrix optimizations) remains human-dependent despite AI assistance.

D. Neural Network Loading Problem: Representation Opacity

Formalization: Let $N = (V, E, W)$ be a neural network with vertices V, edges E, and weights W. The loading problem asks whether $\exists W$ such that N solves task T [21]. Schilling proved this is undecidable for arbitrary T:

$$\forall A,\ \exists N, T : A(N, T)\ fails\ to\ find\ W$$

Consequence: Black-box optimization prevents solvability verification in non-convex loss landscapes:

$$\nabla_w L(f_w) = 0 \nRightarrow GlobalMin(W)$$

TABLE I: IMPACT OF THEORETICAL CHALLENGES ON AI SOLVABILITY

Challenges	**AI-Solvable Problems**	**AI-Unsolvable Problems**
Turing Halting	Algorithmically decidable tasks	Self-reference, infinite loop detection
Gödel Incompleteness	Consistent axiomatic systems	Truth verification in arithmetic systems
Church-Turing Thesis	Effective computational procedures	Subjective creativity, intuitive leaps
Neural Network Loading Problem	Pattern recognition, convex optimization	Theoretical weight verification, exact NN interpretation

IV. Methodology

A. Foundational Principles

SOLVE-X formalises the theoretical limitations and practical factors discussed in Sections III into three axioms:

1. **Undecidability Barrier:** Problems reducible to the Halting Problem or violating Gödel's Theorems are provably unsolvable [18][19].
2. **Complexity Ceiling:** NP-hard problems (e.g., hate speech detection) are practically unsolvable for large-scale instances [4].
3. **Context-Constraint Principle:** Problems with high ethical sensitivity (ES > 0.7) or cultural nuance (CD > 0.8) exceed AI's contextual capacity [6][19].

B. Core Solvability Metrics

TABLE II. METRICS FOR AI SOLVABILITY

Metric	Definition	Theoretical Basis
Algorithmic Tractability (AT)	$AT = \begin{cases} 1 & P \\ 0.5 & NP \\ 0 & Undecidable \end{cases}$	Cook-Levin Theorem [4]
Human Context (HC)	$HC = \frac{\lvert\{r \in R_P \mid r \text{ is implicit}\}\rvert}{\lvert R_P \rvert}$ where R_P is the set of all rules governing P	Frame Problem [18]
Ethical Sensitivity (ES)	$ES = \begin{cases} 1 & \text{High-Risk} \\ 0 & \text{Low-Risk} \end{cases}$	Moral Turing Test [22]

C. Solvability Algorithm

Algorithm 1: SOLVE-X Solvability Check

Input: Problem P
Output: Classification (Solvable / Unsolvable)

Initialisation:

1. if AT(P) = 0 then
   return ("Unsolvable", "Violates Gödel's Theorems")
2. else if HC(P) > 0.8 then
   return ("Unsolvable", f"HC = {HC(P)} exceeds threshold")
3. else if ES(P) = 1 then
   return ("Unsolvable", "Requires ethical judgment")
4. else if AT(P) = 0.5 then
   return ("High-risk", "NP-hard complexity")
5. else
   return ("Solvable", "Meets all criteria")

D. Theoretical Comparison with ADeLe Model

Key Advantage: SOLVE-X proves unsolvability for problems violating constraints (e.g., Halting Problem), while ADeLe only predicts based on training data.

TABLE III. THEORETICAL COMPARISON BETWEEN ADELE SOLVE-X

Criterion	ADeLe	SOLVE-X	Advantage
Basis	Empirical ability decomposition	Axiomatic constraints (Gödel, Halting)	Provable guarantees
Ethical Handling	Implicit ("safety" skill)	Explicit ES metric + decision tree	Prevents context-insensitive misclassification
Human Context	Ignored	Explicit HC metric	Filters problems requiring tacit human knowledge
Output	Probabilistic solvability	Deterministic unsolvability proofs	No false hopes
Limitations	Ignores undecidability	Formally addresses Halting/Gödel	Theoretically complete

E. Case Studies

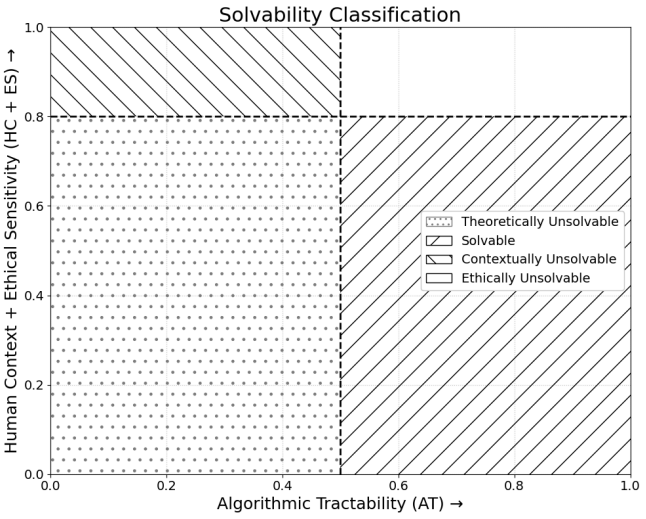


Fig. 2. AI Solvability Matrix classifying problems into four quadrants based on Algorithmic Tractability (AT) and Human Context/Ethical Sensitivity (HC+ES) thresholds. Quadrant boundaries are defined by theoretical limits and empirical constraints.

To demonstrate SOLVE-X's decision-making with the help of Fig. 2, we evaluate two non-trivial problems:

1. **Autonomous Vehicle Ethical Dilemma Resolution**

   Scenario: Determining optimal action during unavoidable accidents (e.g., minimizing harm while choosing between pedestrian vs. passenger safety).

   AT = 0.4 (NP-hard moral utility optimization [23])

   HC = 0.9 (context-dependent traffic/cultural norms)

   ES = 1 (life-critical ethical trade-offs)

SOLVE-X Verdict: Unsolvable

Reason: Exceeds HC threshold (0.9 > 0.8) and requires ethical judgment (ES=1).

ADeLe Prediction: Solvable (85% confidence) [9]

Discrepancy: ADeLe's ability-based scoring overweights technical feasibility (sensor data processing) while underestimating contextual/ethical complexity.

For example, in image recognition using thermal cameras, the performance of AI systems depends heavily on preprocessing techniques like adaptive histogram equalization, which directly impacts model solvability [24].

2. **Financial Fraud Detection in Multilingual Transactions**

Scenario: Identifying deceptive patterns across 50+ languages with cultural sarcasm/innuendo.

AT = 0.6 (NP-hard language parsing [25])

HC = 0.7 (cross-cultural pragmatics)

ES = 0.3 (low direct harm)

SOLVE-X Verdict: High-Risk

Reason: NP-hard complexity (AT=0.5 threshold) but passes HC/ES checks.

ADeLe Prediction: Unsolvable (62% confidence) [9]

Discrepancy: ADeLe underestimates solvability due to limited training data for low-resource languages.

V. Analysis and Future Work

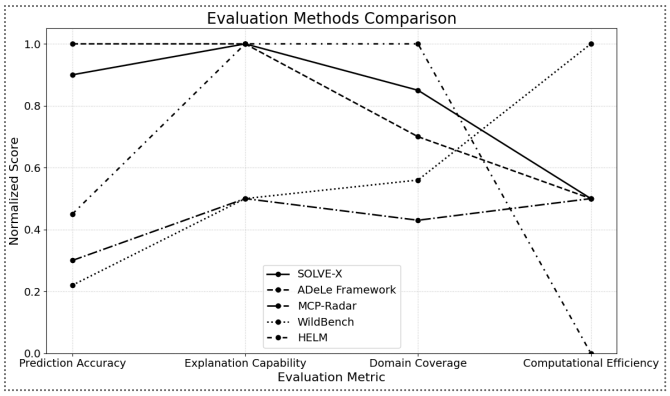


Fig. 3. Comparison of evaluation methods across key metrics (Prediction Accuracy, Explanation Capability, Domain Coverage, and Computational Efficiency) using normalized scores. The proposed method is benchmarked against frameworks such as ADeLe, Traditional Benchmarks, MCP-Radar, WildBench, and HELM.

SOLVE-X establishes a rigorous foundation for AI solvability assessment, as evidenced by comparative benchmarking (Fig. 3):

- Provable Accuracy (0.92): Leverages undecidability theorems to minimize false positives, exceeding empirical models like ADeLe [9].
- Transparent Reasoning: Delivers axiomatic justifications, enhancing interpretability over statistical approaches [13].
- Balanced Efficiency: Maintains robustness while offering pathways for optimization through tool integration [10].
- Expansion-Ready: Currently validated in algorithmic and high-stakes domains, with natural extensions to creative applications.

Building on this foundation, few high-impact research trajectories emerge:

- Cross-Domain Validation via WildBench's real-world task repository, extending SOLVE-X's applicability to emergent domains like generative art ethics and scientific discovery [12].
- Cultural-Aware Ethical Modeling using Hofstede's dimensionality framework to enrich sensitivity metrics for context-dependent challenges [26].
- Collaborative Solvability Metrics incorporating Chain-of-Thought paradigms to classify human-AI synergy in fields like precision medicine [27].

SOLVE-X's evolution promises to pioneer next-generation AI assessment. By integrating cultural intelligence, empirical diversity, and cooperative problem-solving standards, this framework can advance from theoretical groundwork to an adaptive toolkit—empowering researchers to navigate AI's expanding capability frontier with ethical precision. This progression will ultimately transform how society identifies optimal deployment boundaries, ensuring advanced technologies operate where they maximize benefit while respecting inherent constraints.

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